



FAKULTI SAINS DAN SUMBER ALAM
UNIVERSITI MALAYSIA SABAH

CURRICULUM VITAE

Seeking Industrial Training



Name : RACHANA A/P DIPAN
IC/Passport No. : 010629-01-1882
Student No. : BS22110099
Nationality : MALAYSIAN
Address : NO 5, JALAN SS 5A/12, KELANA JAYA 47301 PETALING JAYA, SELANGOR.
Home /Mobile Phone No : +60 18-9855493
E-mail : rachana_dipan_bs22@iluv.ums.edu.my
LinkedIn : www.linkedin.com/in/rachana-dipan-939b23237

• **PERSONAL DETAILS**

Age : 24

Marital Status : Single

Interests : Software Development ,Computer Vision, Machine Learning, AR/VR Development, UI/UX Design, 3D Graphics & VTK, Data Analysis, Creative Design, Diving & Marine Activities

• **ACADEMIC QUALIFICATIONS**

UNIVERSITI MALAYSIA SABAH

Bachelor of Science with Honors (Mathematic with Computer Graphic)
To be awarded in December 2026 [Current CGPA: 3.31]

Major Course

Year 1

- Basic Statistics
- Logic and Problem-Solving Design in Science and Technology
- Calculus
- Data and Analytics in Science and Technology
- Numerical Linear Algebra
- Computer Programming
- Discrete Mathematics

Year 2

- Basic Physics
- Advanced Calculus
- Object Oriented Programming
- Mathematical Cryptography
- Earth Science
- Foundation of Real Analysis
- Data Structure
- Fundamental of Computer Graphics
- Fuzzy Math

Additional Electives

- Philosophy and Contemporary Issues
- Appreciation of Ethics and Civilization
- Fundamentals of Entrepreneurial Acculturation
- Government and People in Southeast Asia
- Media and Society
- Foreign Language Grade 1,2,3 (Mandarin)

Year 3

- Differential Equations
- Geometry Modelling
- Methodology and Scientific Writing
- Database
- Digital Image Processing
- Real Time Graphics
- Numerical Optimization
- Mathematical Modelling and Visualization
- Fundamentals of Machine Learning
- Computer Vision and Application
- Integrated Human-Computer Interface
- Scientific Project I

Year 4

- Scientific Project II
- Artificial Intelligence
- Scientific Data Visualization
- Numerical Method for Modelling

STPM – SEKOLAH MENENGAH KEBANGSAAN JOHOR JAYA (1) (2020-2022)

INFORMATION AND COMMUNICATIONS TECHNOLOGY – CGPA: 2.42

SPM – SEKOLAH MENENGAH KEBANGSAAN SULTAN ABDUL SAMAD (2014-2018)

SCIENCE STREAM (COMPUTER SCIENCE) – 3B+, 2B, 2C+

• WORKING EXPERIENCES

Store Assistant – 7-Eleven Sdn. Bhd., Malaysia

Nov 2018 – Mar 2019

- a. Managed daily cashier operations including cash and e-payment transactions.
- b. Maintained and restocked inventory.
- c. Handled customer service and assisted with product arrangement.

• RESEARCH EXPERIENCES

Final Year Project : Confusion Lines, Background Dots, and Character Distortion Robust Text-Based CAPTCHA Detection and Similar Character Disambiguation Using Adaptive Preprocessing, CRNN + Attention, Center Loss.

Project Supervisor: Ts. Rechard Lee (rechard@ums.edu.my)

• AWARDS

- Dean's List – Semester 2 (2023/2024)
- Dean's List – Semester 2 (2024/2025)
- Champion — Co-Curricular Carnival of the Year (Theatre Performance)

• CO-CURRICULUM ACTIVITIES

- Member – MCG Club (Mathematics with Computer Graphics)
- Member – Indian Community Club (SEKWIM)
- Member – RACUMS Club
- Committee Member – Mathematics Leadership Camp 2025
- Member – Friends of Meraki Daat (NGO)
- Swimming/Diving and marine-related activities
- Gaming and creative digital hobbies

• LANGUAGE PROFICIENCY

- WRITTEN
 - English – Excellent
 - Malay – Excellent
 - Tamil – Excellent
 - Mandarin – Basic
- SPOKEN
 - English – Excellent
 - Malay – Excellent
 - Tamil – Excellent
 - Mandarin – Basic

• COMPUTER SKILLS

Programming & Development	Software & Creative Tools	Mathematics, Analysis & Productivity Tools
• Python • C, C++, C# • Java • HTML, CSS • SQL (MySQL) • R Studio • Software Development • Data Analysis	• Unity (C#) – Intermediate • Blender (VTK) – Basic • Figma (UI/UX Design) – Intermediate • Canva (Infographics & Layout) • CapCut (Videography & Editing)	• MATLAB (Basic) • Lingo (Optimization) • Microsoft Office (Word, Excel, PowerPoint, Teams) • Google Workspace (Docs, Forms, Drive)

• LABORATORY SKILLS

- Image Processing (filtering, histogram, thresholding, edge detection)
- Feature Extraction (Harris, Shi-Tomasi, SIFT implemented from scratch)
- Deep Learning model training (TensorFlow & PyTorch)
- Machine Learning workflows (SVM, CNN, evaluation metrics)
- Dataset preparation, preprocessing & augmentation
- 3D modelling, rendering & visualization using VTK/Blender
- Real-time GUI development for CV and ML applications
- AR/VR prototyping (Unity + Vuforia)
- Performance testing (SSIM, PSNR, confusion matrix, precision/recall)
- Scientific graphics & visualization methods

• REFERENCES

Ts. Rechard Lee

Final Year Project Supervisor,
Head of Mathematics and Computer Graphics Programme
Faculty of Science and Technology (FST)
Universiti Malaysia Sabah (UMS)
rechard@ums.edu.my
☎ 103651 / (+60)14 – 650 1664

Professor Dr. Abdullah Bin Bade

Professor,
Faculty of Science and Technology (FST)
Universiti Malaysia Sabah (UMS)
abb@ums.edu.my
☎ 102748 / (+60)13-7204450